Journal of Biomechanics 196 (2026) 113148

Contents lists available at ScienceDirect

Journal of Biomechanics

journal homepage: www.elsevier.com/locate/jbiomech

Functional and anatomical characteristics of the heel fat pad based on different sports events

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ARTICLE INFO

Keywords:
Heel fat pad
Microchamber layer
Macrochamber layer
Ultrasound
Stiffness

ABSTRACT

This study investigated the thickness and stiffness of the heel fat pad among male university athletes involved in sports with different performance characteristics: with shoes, barefoot, and in water. Participants included 34 male athletes—12 judo players (barefoot competition), 10 American football players (shoe competition), and 12 swimmers (water-based)—as well as 12 male non-athlete controls. Using ultrasonography, we measured the thickness of the shallow and deep layers of the heel fat pad under 0%, 50%, and 100% loading conditions, along with their respective rates of change. Stiffness (elasticity) of both layers was also assessed. At 0% load, both the shallow and deep layers were thickest in swimmers and thinner in judo and American football athletes. In the shallow layer, swimmers showed the greatest rate of change in thickness with loading. In the deep layer, significant changes were found in the judo and swimmer groups compared to the American football group, and in swimmers compared to controls. Elasticity in the shallow layer was lower in American football players than in the judo and swimmer groups. In the deep layer, judo athletes had the lowest elasticity, whereas swimmers had the highest. These findings indicate that the heel fat pad's morphology and mechanical properties differ across sports disciplines, likely reflecting adaptations to sport-specific loading patterns.

1. Introduction

The heel (calcaneal) fat pad has a cup structure that encompasses the bottom, medial, lateral, and posterior sides of the calcaneus (De Clercq et al., 1994: Campanelli et al., 2011; Fontanella et al., 2016; Pettenuzzo et al., 2023). Its main function is to absorb shock while walking and running and its dysfunction may lead to heel pain (Hsu et al., 2009; Kumai and Benjamin , 2002). In patients with heel pain, degeneration and destruction of the elastic fibers forming the heel fat pad, reduction of healthy adipose tissue, and outflow of lipid droplets have been observed, resulting in heel thinning (Jørgensen and Bojsen-Møller, 1989). Consequently, morphological changes (thickness changes) and functional changes (elastic changes) occur in the heel fat pad, decreasing shock absorption ability and increasing compressive stress on the calcaneus (Kwan et al., 2010; Chao et al., 2010; Belhan et al., 2019; Maemichi et al., 2020; Kwak et al., 2020; Maemichi et al., 2024). Changes in the form and function of the fat pad have been speculated to cause heel pain.

The biomechanical behavior of the plantar fat pad, including its viscoelastic properties and its response to various loading conditions, has been extensively studied in healthy and degenerative foot conditions (Fontanella et al., 2016; Pettenuzzo et al., 2023). Furthermore, studies have investigated effects of aging and systemic diseases, such as diabetes, on the stiffness and thickness of the plantar soft tissues (Kwan et al., 2010; Chao et al., 2010; Kwak et al., 2020).

Each competitive sport has unique movement styles, and the amount of load placed on the body differs. The sole, particularly, is directly impacted by these sports. Accordingly, significant differences may exist in the form and functional characteristics of the heel fat pad depending on the sport. The height of the foot arch reportedly differs depending on the sport (Kobayashi et al., 2020). Focusing on sports competitions, the morphology and functional characteristics of heel fat pads during each

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https://doi.org/10.1016/j.jbiomech.2026.113148
Accepted 1 January 2026
Available online 7 January 2026
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sport have not been evaluated previously. For competitive athletes, pain in the soles of their feet is directly linked to performance decline and is a serious condition that can affect their lives.

Body weight is strongly related to heel fat pad thickness (Kuhs et al., 1949; Maemichi et al., 2024). Furthermore, as the body weight and load increase, the rate of change in the thickness of the heel fat pad changes when a load is applied (Maemichi et al., 2024; Matsumoto et al., 2025). Therefore, the morphology and functional characteristics of heel fat pads may change depending on the amount of load applied.

While previous studies have demonstrated that aging, body weight, and systemic diseases affect the morphology and mechanical properties of the heel fat pad, little attention has been paid to the influence of sport-specific loading environments. In particular, quantitative comparisons of heel fat pad thickness, load-dependent deformation, and stiffness among athletes exposed to markedly different contact conditions (barefoot, shoe-based, and aquatic sports) have not been reported. The present study was therefore designed to clarify whether distinct sport-specific adaptations of the heel fat pad can be detected using combined morphological and mechanical assessments. The observed differences among sports in both thickness and stiffness support the relevance of evaluating heel fat pad behavior in relation to sport-specific loading characteristics.

2. Methods

2.1. Participants

The participants were 34 male university athletes who participated in university athletic club activities (12 judo players (JD) (barefoot sport), 10 American football players (AF) (shoe-based sport), and 12 swimmers (SW). We also included 12 male university students in the control group (CON) who did not perform any specific competitive habits and had not participated in any sports activities in the past year. All participants participated in university athletic club activities and competed for more than 5 h a day.

The exclusion criteria were participants (1) who engaged in strenuous physical activity on the day of measurement; (2) who consumed alcohol within the past 24 h; (3) had a history of foot or ankle surgery or major trauma; (4) with heel pain or orthopedic injuries to the ankle or foot, such as ligament injuries; (5) with rheumatic diseases such as osteoarthritis, gout, and rheumatoid arthritis; and (6) with systemic diseases such as diabetes and/or connective tissue disorders.

The purpose of the study, measurement methods, and ethical considerations were explained to the participants in advance, and informed consent was obtained. This study was approved by the Ethics Review Committee for Research Involving Human Subjects (approval number 2020–228).

2.2. Protocol

2.2.1. Load measuring instruments

Polymethylpentene resin plates (PMP) plates were used for evaluation. The usefulness of PMP for measuring loads on the heel fat pads has been reported (Matsumoto et al., 2022; Maemichi et al., 2024). We prepared an evaluation table with a 350-mm square, a 20-mm thick polycarbonate resin top plate on a 400-mm high four-leg iron frame, and a scale stand to measure the load. There was a cross hole in the top plate, allowing the ultrasound probe to capture images from the bottom. A water tank composed of a 5-mm thick PMP plate measuring 90 mm in height, 350 mm in length, and 155 mm in width was fixed in the cross hole cut out in the top plate, and water was stored in the tank to a height of 10 mm. Since the average foot axis angle during walking was approximately 7° from the sagittal plane, the two evaluation tables were placed so that they were separated by 7° from the sagittal plane. The lower edge of the calcaneus prominence was detected from the sole in the long-axis direction through the PMP board, and the second metatarsal and calcaneus protuberance were placed on the straight line drawn on the top plate of the load-measuring device in the water tank. The participant was allowed to place his foot inside.

2.2.2. Ultrasound measurement

We used an ultrasound imaging device (Applioα, Canon, JPN) with a 14 MHz high-frequency linear probe. The thickness of the heel fat pad was measured at three points—when no load was applied (0% load), standing on both legs (50% load), and standing on one leg (100% load)—and was divided into shallow and deep layers. The load thickness change rates between 0 − 50%, 50 − 100%, and 0 − 100% were calculated. The formula used to calculate the rate of change is presented in Table 1. The elasticity of the heel fat pad was measured in the prone position and divided into shallow and deep layers.

Shallow and deep layers were classified as the macrochamber layer (MAC layer) (including the plantar aponeurosis and MACs) from the most convex part of the calcaneal protuberance to the bottom of the fibrous septum (FS), according to previous reports (Matsumoto et al., 2023; Maemichi et al., 2024). The area from the bottom of the FS to the skin was defined as the microchamber layer (MIC layer) (including MICs). The intra-examiner reliability of the heel fat pad thickness measurements at 0%, 50%, and 100% load was determined using intraclass coefficient (ICC) one-way random effects (model 1), single measurement, and agreement. At both 50% and 100% loads, it was $\geq$ 0.89 in both shallow and deep layers. Intra-examiner reliability for measuring heel fat pad elasticity was $\geq$ 0.90 for the shallow and deep layers using ICC one-way random effects (model II), single measurement, and agreement.

2.2.3. Morphological measurement (heel fat pad thickness)

For nonloading (0% load), the ankle was in a neutral-plantar dorsiflexion position, the knee and hip were in a 90° flexion position, and both hands were placed on the anterior superior iliac spine in a sitting position. In the two-legged standing position (50% load), the participants stood on a load-measuring device shoulder-width apart so that their weight was evenly distributed on both sides. When standing on one leg (100% load), the pelvis and trunk were moved to the right so that the ankle, knee, and hip joints were in a straight line to ensure that the weight was applied between the medial and lateral malleoli. Once this was confirmed, the patient's left foot was lifted off the ground by a few millimeters, and the patient was placed in a position where the entire body weight was placed on the right foot. The right foot was used for measurement, and the load on the right foot was adjusted based on the value of the left foot scale.

Unlike the examiner performing the ultrasound measurements, a second examiner observed the load and instructed the participant to maintain that load. Additionally, when standing (50% load) and at full load (100% load), the posture was evaluated using a camera from the side to ensure that the participant did not lean forward and that the heel remained in contact with the surface. A mark was placed at eye level during the measurement, and participants were instructed to keep their gaze on the mark throughout. Furthermore, participants were verbally guided to apply weight to the heel. The participants maintained the required posture and weight distribution until stabilization, which typically required approximately 5 s. Ultrasound images were acquired, ensuring that the measured heel fat pad thickness reflected the equilibrium state under each loading condition. Three ultrasound images

Table 1
Calculation formula for evaluating the function of the heel fat pad.

Rate of change (MIC layer)	=	Amount of change (MIC layer) / 0% loading (MIC layer)	×	100
Rate of change (MAC layer)	=	Amount of change (MAC layer) / 0% loading (MAC layer)	×	100

MIC, microchamber; MAC, macrochamber.

were obtained at each measurement point, and the average values of the three images were used.

2.2.4. Mechanical measurements (heel fat pad stiffness)

Stiffness was measured using shear-wave elastography (SWE). Measurements were performed in the prone position, with the knee joint slightly flexed and the ankle plantar dorsiflexed at 0°. To evaluate tissue stiffness under unloaded conditions, SWE measurements were performed before applying body weight. The same procedure as described above for image acquisition was used to ensure stable and reproducible measurements. SWE can be used to evaluate tissue stiffness quantitatively. In principle, the focused tissue is displaced backward by sending acoustically reflected pressure from the ultrasound probe, and the restoring force of the tissue propagates laterally, generating shear waves. The relationship between tissue hardness (E) and the shear-wave propagation velocity (c) can be expressed as $E = 3\rho c2$ (ρ is tissue density). The shear wave propagation speed was high in the hard tissues and low in the soft tissues. Using this velocity, it is possible to calculate the degree of tissue hardness. SWE is a color stiffness map superimposed on a B-mode image, with hard objects in red and soft objects in blue (Telfer et al., 2014; Tietze., 1982). For quantitative stiffness measurements, the built-in software displayed the maximum, minimum, standard deviation, and average shear rates in kilopascals (kPa) or speed units (m/s) and in this study, it was unified as m/s. Since the ultrasound system used in this research is a commercialized system, it was difficult to change the shape of the region of interest (ROI); therefore, all analyses were performed in circles.

Regarding the hardness measurement of the heel fat pad for the shallow layer, three ROIs were taken from one still image; the average value was taken as the average value of one image, and the average value of the three images was adopted. For the deep layer, one ROI was taken from each image, and the average value of the three images was used (McCarthy and Gorecki, 1979) (Fig. 1).

2.3. Statistical analysis

Statistical analyses were performed using IBM SPSS Statistics (ver.26, IBM Corp., NY, USA). The Shapiro − Wilk test was performed on each measurement item to confirm normality. Comparisons between the three points of nonloading (0% load), standing on both legs (50% load), and standing on one leg (100% load) in the shallow and deep layers of the heel fat pad were performed twice for each sport. A configuration analysis of variance (ANOVA) was performed. Similarly, a two-way ANOVA was used to determine the rate of change in the load thickness of the shallow and deep heel fat pads for each sport. One-way ANOVA was used to measure the stiffness in the shallow and deep layers of the heel fat pad. The significance level was set at 0.05.

3. Results

3.1. Participants

There was no significant difference in age between participants of different sports, but the participants in the CON group were significantly older than those in the three sports groups. There were no significant differences in height among the four groups. Bodyweights were significantly higher in JD and AF groups than in SW and CON groups. The SW group had a significantly longer competition history than JD and AF groups, and the JD group had a significantly longer competition history than the AF group (Table 2).

3.2. Morphological measurement (heel fat pad thickness)

The deep layer of the heel fat pad was significantly thicker in JD, AF, and SW groups than in the CON group at 0% load. No significant differences were observed between the four groups at a 50% load. At 100% load, the thickness in the AF group was significantly greater than that in the SW group.

The superficial layer of the heel fat pad was significantly thicker in the SW group than in JD, AF, and CON groups at a 0% load. At a 50% load, the thickness of the SW group was significantly greater than those of AF and CON groups. No significant differences were observed between the four groups at a 100% load.

The deep layer of the heel fat pad was significantly thicker in the SW group than in JD, AF, and CON groups at a 0% load. At 50% and 100% loads, AF and CON groups were significantly thicker than the JD group. The superficial layer of the heel fat pad was significantly thicker in the SW group than in JD, AF, and CON groups at a 0% load. Additionally,

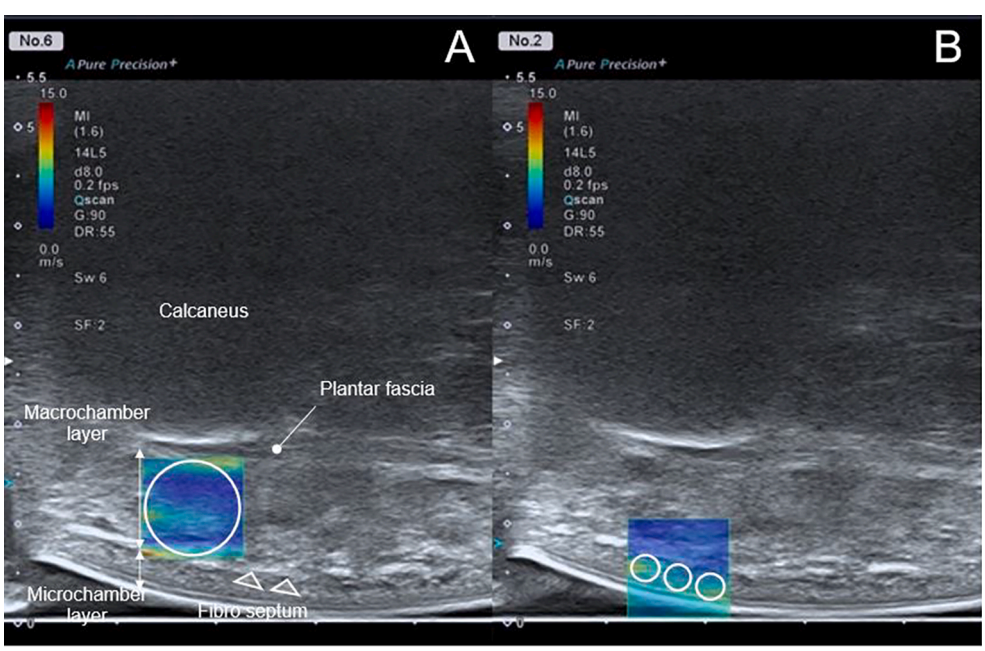


Fig. 1. Measurement of the shear modulus of the heel pad in the long axis (A) Microchamber layer (three side-by-side regions of interest were selected to measure the shear modulus). (B) Macrochamber layer.

Table 2
Physical characteristics of participants.

	n	Age (years)			Height (cm)			Body mass (kg)			Athletic career (years)		
JD	12	21.0[#]	±	2.0	175.5	±	6.3	92.4[†,#]	±	11.6	13.6 [*]	±	2.4
AF	10	20.7[#]	±	1.1	174.3	±	3.3	84.1[†,#]	±	12.7	5.6[†]	±	1.6
SW	12	19.7[#]	±	1.2	175.3	±	4.3	70.0	±	5.3	15.6	±	1.8
CON	12	24.6	±	1.9	170.6	±	4.0	63.4	±	7.5	–		

Data are shown as means and standard deviations: mean ± SD.
JD, judo group; AF, American football group; SW, swimming group; CON, control group.
[*] : $p < 0.05$ Compared with AF.
[†] : $p < 0.05$ Compared with SW.
[#] : $p < 0.05$ Compared with CON

the CON group was significantly thicker than JD and AF groups. At 50% and 100% loads, the SW group was significantly thicker than JD and AF groups. Additionally, the CON group was significantly thicker than JD and AF groups (Fig. 2).

The rate of change in the load thickness of the deep layer of the heel fat pad was significantly greater in JD and SW groups than in the AF group (from 0 – 50%). At 50 – 100%, the change was significantly greater in the SW group than in the AF group. From 0 – 100%, the change was significantly greater in the SW group than in AF and CON groups. Furthermore, this change was significantly greater in the JD group than in the AF group. The rate of change in the load thickness of the superficial layer of the heel fat pad was significantly greater in the SW group than in JD, AF, and CON groups at 0 – 50%, 50 – 100%, and 0 – 100% (Fig. 3).

A significant decrease in the load thickness of the deep layer of the heel fat pad was observed in all groups at 0%, 50%, and 100% loads. For the superficial layer, a significant decrease was seen only in the SW group at all three load conditions. No significant changes were observed in the other three groups. After correcting for body weight, a significant decrease in the thickness of the deep heel fat pad layer under 0%, 50%, and 100% loads was observed in all groups. For the superficial layer, a significant decrease was seen only in the SW group across all load levels. No significant changes were observed in the other three groups.

In all groups, the rate of change in the deep layer of the heel fat pad was significantly greater between 0 – 50% load than between 50 – 100% load. For the superficial layer, this difference was only observed in the SW group; no significant differences were observed between the other groups (Fig. 4).

3.3. Mechanical measurement (heel fat pad stiffness)

The stiffness of the deep layer of the heel fat pad was significantly higher in the AF group than in JD and SW groups. The stiffness of the superficial layer of the heel fat pad was significantly higher in the JD group than in AF, SW, and CON groups. These values were significantly higher in AF and CON groups than in the SW group (Fig. 5).

4. Discussion

This study examined the heel fat pad of university athletes in three sports with different characteristics: sports played with shoes, barefoot, and in water. We focused on stiffness and examined its morphological and functional characteristics. As body weight is reportedly strongly related to the thickness of the heel fat pad when unloaded and to the rate of change in the deep layer, and since the JD and AF groups were significantly heavier than SW and CON groups, we analyzed weight-corrected data by dividing heel pad thickness by body weight.

At a 0% load, the SW group had the thickest pad compared to the other groups. The CON group (no participation in competitions), was thicker than JD and AF groups. Similar results were obtained for the thickness of the deep layer of the heel fat pad. Additionally, in the 0 – 100% load thickness change rate in the shallow layer and that in the deep layer, the SW group and the JD group, respectively, had a significantly larger load thickness change rates than the other groups. Furthermore, the results showed that the rate of change in load thickness was significantly greater in the SW group than in the AF group and in the SW group than in the CON group.

As the SW group played underwater sports, the load on the heel and the frequency of contact with the ground were expected to be lower in the SW group than those in the JD and AF groups, as well as in the CON group; the degeneration of the heel fat pad due to load was also expected to be lower than that in normal healthy people of the same age. Supporting the reliability of our SWE measurements, a previous study using SWE has reported quantitative stiffness values of the heel fat pad (Lin CY et al., 2017). A study on plantar heel pain (Lin CY et al., 2015) reported that, in their control group without heel pain, the MIC, MAC, and whole layers exhibited mean stiffness values of 60.9 ± 11.4 kPa, 26.3 ± 6.1 kPa, and 31.8 ± 6.3 kPa, respectively. The stiffness values obtained in the CON group of this study under unloaded conditions were comparable to these previously reported control values. This alignment suggests that the mechanical characteristics measured in our study are consistent with established reference data, supporting the validity of our baseline SWE measurements. It is possible that tissue deformation did not occur or that the degree of deformation was delayed or minimal owing to the small amount of load. Additionally, the participants competed in water for > 5h/day with long periods in water (>20% of the day). Keratin, a protein present in the stratum corneum (the surface layer of the skin), has been suggested to retain water, causing a temporary increase in the moisture content of the stratum corneum and resulting in a temporary increase in volume. Consequently, we infer that the shallow layer at a 0% load was thicker in the SW group, and the rate of change in the load thickness of the shallow layer was significantly larger in the SW group. Moreover, previous studies have reported normative thickness values for the heel fat pad in healthy adults, with mean thicknesses of approximately 14 mm in males and 12 mm in females (Morag et al., 1997; Turgut et al., 1999; Maemichi et al., 2020). In our study, the sum of MIC and MAC thicknesses under unloaded conditions in the CON group were 13–15 mm, which is consistent with these established normative values. This alignment supports the reliability of our morphological measurements and strengthens the interpretation of sport-specific differences observed in the athlete groups. Compared with the CON group, JD and AF groups had a higher amount of load on the heel and more frequent contact with the ground. We predicted that transient overload and load that was not applied in daily life was applied to the heel, suggesting that changes such as the outflow of lipid droplets or degeneration of the septum layer, occur. Additionally, although there was no difference in the thickness of the deep layer at a 0% load, there was a difference between JD and AF groups in the rate of change in the thickness of the deep layer with load, with a marked change observed in the JD group. De Clercq et al. (Wearing et al., 2009) reported that the change in heel fat pad thickness under load was reduced by half when wearing shoes compared to being barefoot. The rate of change of the load on the heel fat pad reportedly differs depending on the shape of the insole when shoes are worn (Wu et al., 2018). Based on these findings,

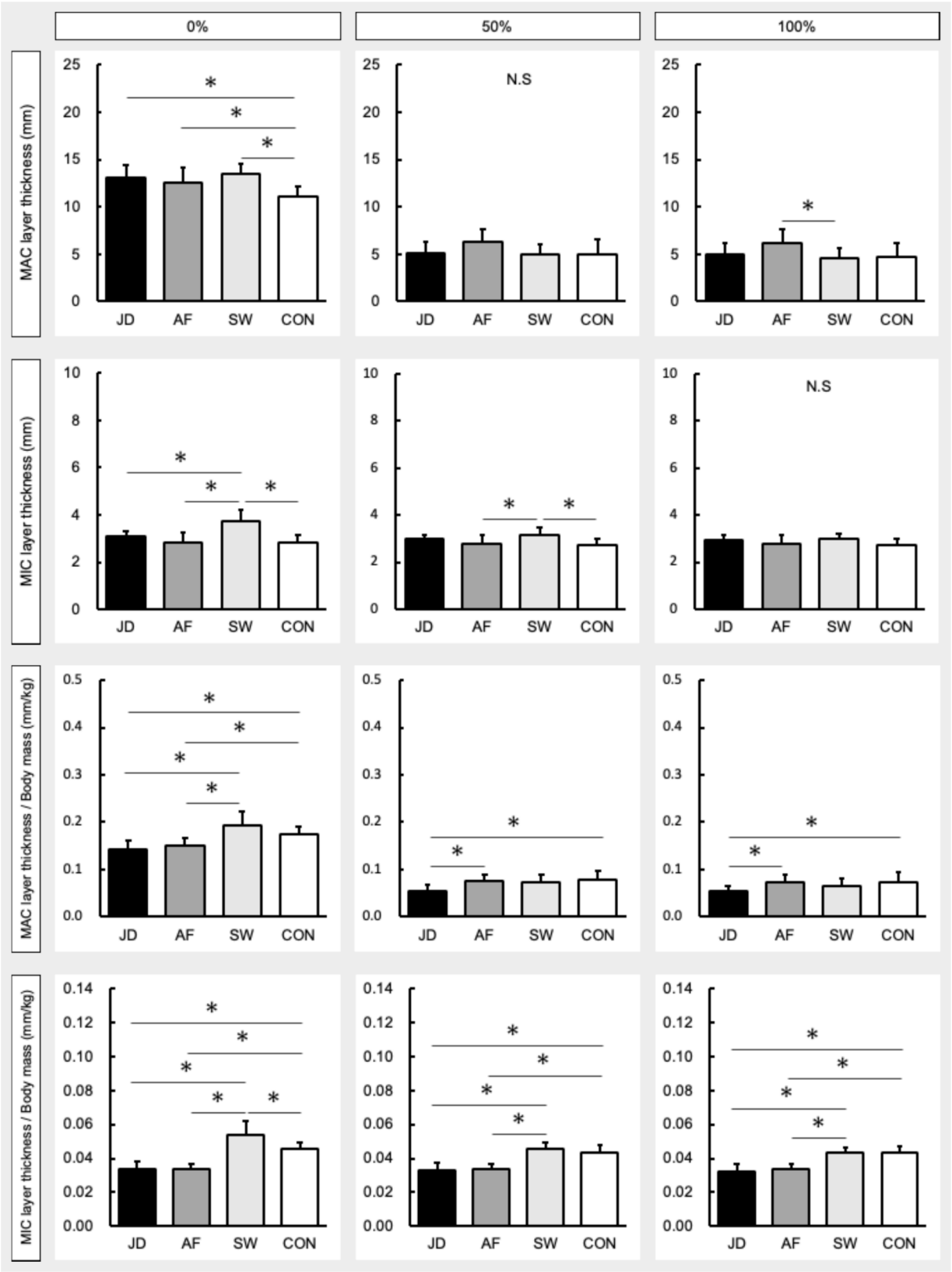


Fig. 2. Comparison of the thickness between the macrochamber and microchamber layers. Comparison of macrochamber and microchamber layer thickness (body mass correction). *Significantly different from other groups ($P < 0.05$).

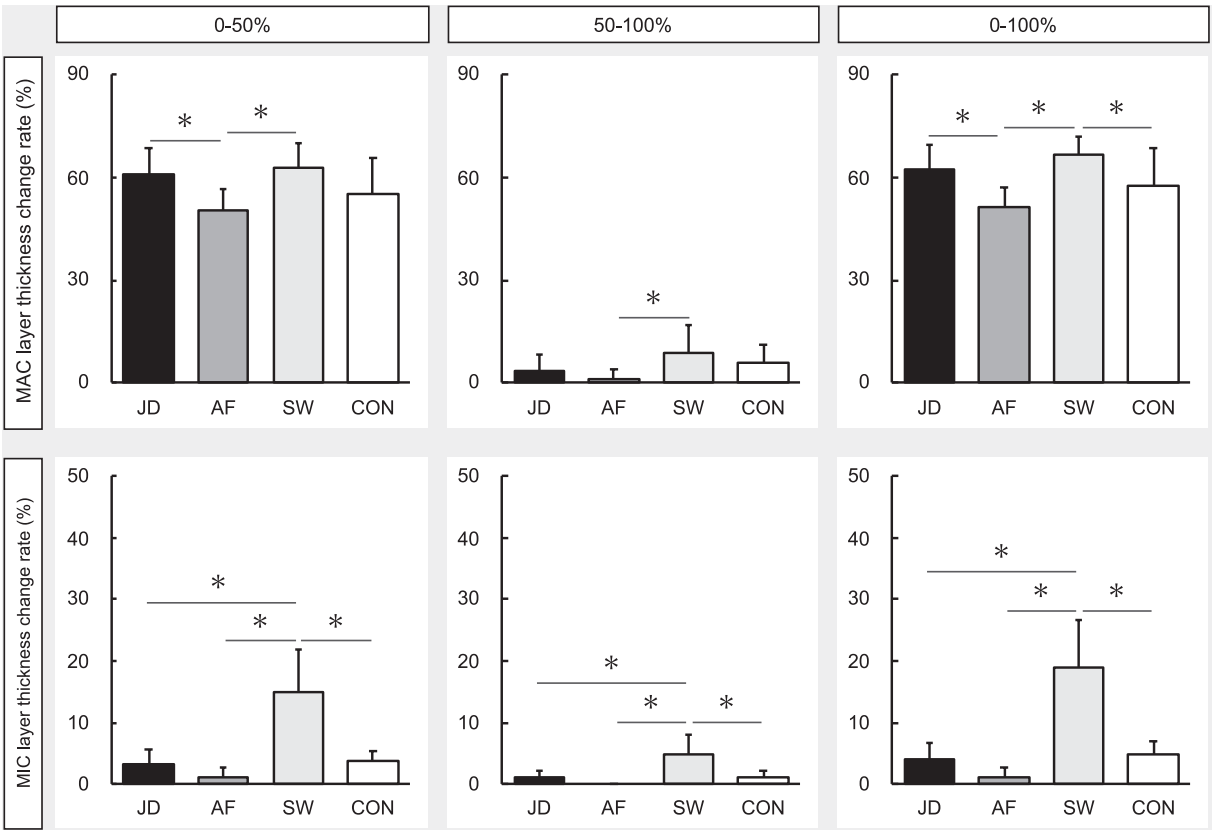


Fig. 3. Comparison of the thickness between the macrochamber and microchamber layers. Comparison of macrochamber and microchamber layer thickness (body mass correction). *Significantly different from other groups ($P < 0.05$).

the JD group, including athletes who competed barefoot, was predicted to be subjected to more stress on their heels than the AF group, who wore shoes, and that qualitative changes would occur within the heel fat pad. These findings are speculative, and further investigations are required.

The JD group had the highest stiffness values for the shallow layer, whereas the SW group had the lowest. Conversely, the AF group had higher stiffness values than the JD and SW groups in the deep layers. These differences may primarily relate to shoe-wearing, differences in sport characteristics between land and water, differences in load amount and frequency, and water retention. Furthermore, the effect of sport characteristics (land vs. water) on the epidermis was also considered a factor. The JD group competed barefoot on a tatami, which applied direct pressure and friction to the soles. Direct stimulation of the plantar surface has been shown to thicken the surface layer of the stratum corneum. Such changes have been reported in judo athletes and may have occurred in this study's participants. Contrastingly, in the JD group, the deep layer tended to be soft, and a similar tendency was observed for the rate of change in thickness due to load. This may be attributable to internal structural damage from repetitive loading.

Although it is difficult to derive the following conclusions from the results of this study, the deep stiffness measurements were similar in the JD and SW groups, healthy adipose tissue morphology may be preserved by minimizing load. Overall, the consistency of stiffness and thickness values with previous literature reinforces the methodological robustness of this study. Furthermore, the identification of sport-specific differences in heel fat pad morphology and mechanical behavior may contribute to the understanding of sport-related adaptations of plantar soft tissues. These findings provide an important foundation for future investigations aimed at identifying risk factors for overuse injuries and developing targeted preventive or rehabilitative strategies tailored to sport-specific loading environments. Even with similar stiffness, the qualitative internal structure and degeneration mechanism are expected to differ, such as in cases where the stiffness is low or where there is degeneration resulting from adipose tissue breakdown owing to overload. When interpreting stiffness, participant's background should be carefully considered.

Although this study suggests that heel fat pad morphology and function differs depending on the sport, other influencing factors cannot be ruled out. Therefore, it cannot be concluded that the findings are solely sport-related, and further research is required.

4.1. Limitations

First, the competitive history of the participants was inconsistent between the groups. Second, the timing of the measurements could not be unified between and within each group. Furthermore, it is unclear whether changes in morphology and function measured by ultrasound reflect histological changes. In future studies, we believe useful information will be acquired by observing locally collected tissues from living organisms using electron and optical microscopes.

5. Conclusions

Our findings revealed that at a 0% load, the superficial and deep layers of the heel fat pad were the thickest in the SW group and thinnest in the JD and AF groups. The greatest change in the thickness of the superficial layer of the heel fat pad was observed in the SW group. The rate of change in the thickness of the deep layer of the heel fat pad was significant in JD and SW groups compared to the AF group and in the SW group compared to the CON group. The stiffness of the superficial layer of the heel fat pad was lower in the AF group than in the JD and SW groups. The stiffness of the deep heel fat layer was both low and high in the JD group and was highest in the SW group.

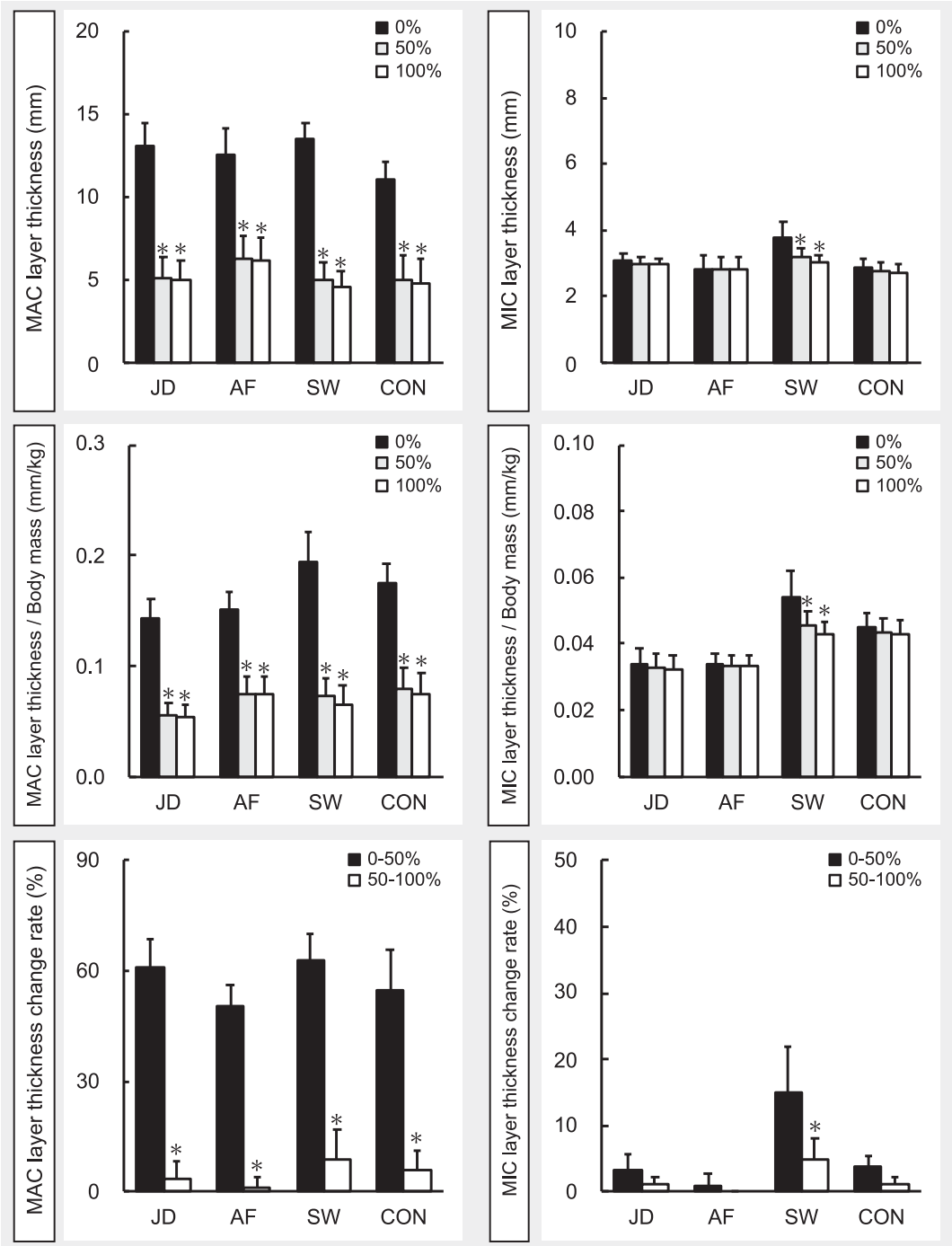


Fig. 4. Comparison of the macrochamber and microchamber layers between loading change processes. Comparison of the macrochamber and microchamber layers between loading change processes (body mass correction). Comparison of the loading change rate in the macrochamber and microchamber layers. *Significantly difference between groups for each loading condition (0%, 50%, 100%) or for the change in thickness between loading intervals (0–50%, 50–100%).

Author contributions:

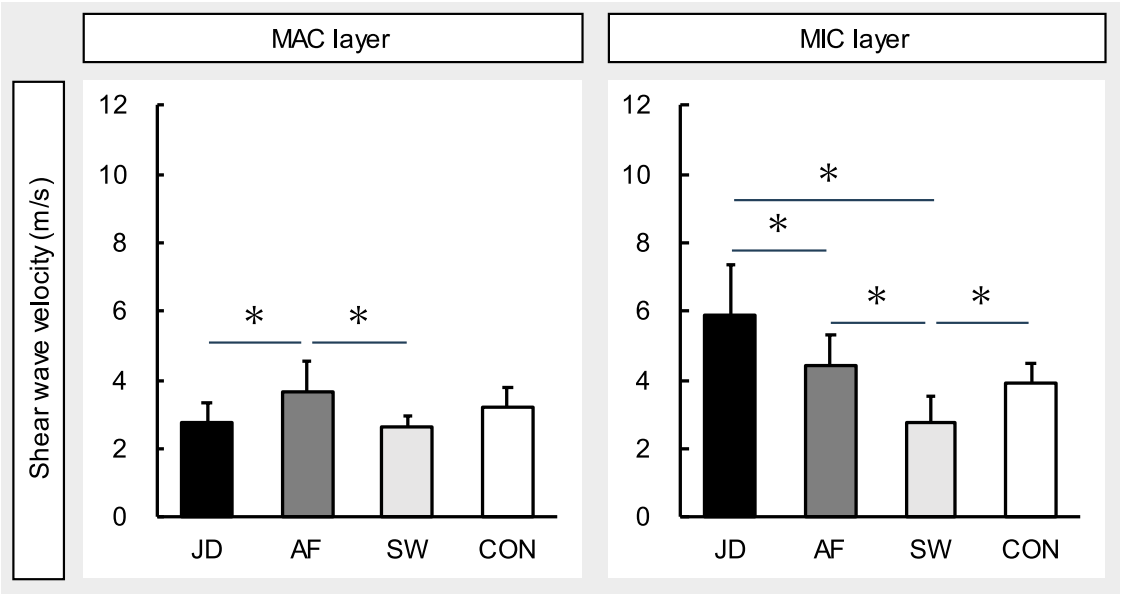
TM was involved in the conception and design of the study; acquisition of approval from the institutional review board of facilities; data acquisition, analysis, and interpretation; and drafting and writing of the article. MM and TO contributed to the conception and design of the study and revised the manuscript critically for important intellectual content. TO and TT were responsible for data acquisition, analysis, and interpretation. TT was responsible for critically revising the paper for important intellectual content. TK advised on all processes in the preparation of the paper and was involved in the final approval of the version to be submitted. All authors read and approved the final manuscript.

7. Data statement:

Please contact the author for data requests.

Funding:

This study was supported by JSPS KAKENHI Grant Numbers 21K09215 and 22K21250.



(A) Macrochamber layer. (B) Microchamber layer. * Significantly different from other groups (P < .05).

Fig. 5. Comparison of the shear wave velocities (m/s) of the macrochamber and microchamber layers. (A) Macrochamber layer. (B) Microchamber layer. *Significantly different from other groups (P < 0.05).

CRediT authorship contribution statement

Toshihiro Maemichi: Writing – review & editing, Writing – original draft, Visualization, Resources, Methodology, Investigation, Data curation, Conceptualization. **Masatomo Matsumoto:** Writing – review & editing, Methodology, Data curation, Conceptualization. **Takumi Okunuki:** Writing – review & editing, Methodology, Data curation, Conceptualization. **Toshiharu Tsutsui:** Writing – review & editing, Methodology, Data curation, Conceptualization. **Tsukasa Kumai:** Writing – review & editing, Supervision, Project administration, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

The authors would like to thank all participants who participated in the measurements. We affirm that the results of this study are presented with clarity, honesty, and integrity, adhering strictly to standards of scientific rigor and ethics.

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